

Bin Duan, PhD

AI · Computer Vision · Multimodal Learning · Biomedical Image Processing

Pathology Faculty Candidate in **Comparative Disease Computation & AI** · College of Veterinary Medicine

PhD, Computer Science, Illinois Tech, Chicago · **Postdoc**, Cell & Developmental Biology, University of Michigan, Ann Arbor



Research program: Comparative pathology foundation models for disease modeling across species and modalities
Companion animals develop the same spontaneous diseases as people, yet veterinary pathology has almost none of the AI built for human medicine. I build AI that learns one shared representation of disease across species, delivering **predictive**, **multimodal foundation-model** methods to veterinary patients first, in tools pathologists use.

THRUST 1

Comparative foundation models

Learning a shared representation, as the predictive engine for different diagnostic tasks.

THRUST 2

Multimodal data integration

Fuse different imaging, molecular, spatial, and clinical modalities into one unified, multimodal patient profile.

THRUST 3

Translation

Deployable, expert-in-the-loop tools for routine clinical or research use in the diagnostic sign-out workflow.

YEARS 1-2

- Canine-human foundation model
- OnePath pilot in sign-out
- First PhD students recruited
- NSF AI Core / CISE submitted

YEARS 3-4

- Open-weight cross-species foundation model
- Audited multimodal benchmark
- Multiple modalities data fusion
- NIH/NCI and USDA/NIFA proposals

YEAR 5

- Expert-in-the-loop tools validated in clinical sign-out / research routine
- First PhD students graduating
- Offer shared digital pathology services to partner clinics & labs

Collaboration plan

Data curation >> Model development >> Deployment

I **lead** methods, data, and platform; **partners bring** cases, data, and validation. Expert verification closes the loop.

- **Disease-model partners:** organoids, canine genetics & cancer genomics, human comparator cohorts.
- **Digital-pathology partners:** spatial transcriptomics, AI in the sign-out workflow, cross-species & wildlife breadth.
- **Computational partners:** foundation-model pretraining, computer vision, calibration & statistical rigor.

Fit + funding

Pathology + Institute of AI + Precision One Health

Bridges the imaging-AI gap in comparative disease computation, from foundation model to diagnostic service.

PI-LED NSF AI Core · CISE · CAREER · NIH R21

COLLABORATIVE NIH/NCI (ICDC) · USDA NIFA (One Health)

Selected work

- *Towards Saner Deep Image Registration*, **ICCV 2023**
Reliable cross-species, cross-modality alignment.
- *Artifact-Minimized High-Ratio Lossy Image Compression with Preserved Analysis Fidelity*,
Nature Communications · under revision Petascale images accessible without losing analysis fidelity.
- *OnePath*, **research prototype** Open-weight, multimodal, cross-species copilot; the working core of all three thrusts and an early prototype of the diagnostic and research service.

Teaching

Literacy, not just use: students learn **when NOT to trust a model**, running and validating methods against domain shift and noisy labels. Proposed AI-Pathology course **Computational Comparative Pathology**, designed for different levels, contributing to UGA's strategic AI degree programs growth.

RESEARCH VISION

PRIOR WORK

TEACHING

Scan and explore



► OnePath demo



Collaboration plan



Image registration



Image compression



Collaborative system



Course syllabus



Sample lecture slides

DIGITAL PATHOLOGY

MULTIMODAL DATA INTEGRATION

PREDICTIVE MODELS